

7 (a) (i) Where the 2 waves meet in phase along AB,

B1

constructive interference occurs and the two amplitudes add up to produce a maximum resultant amplitude, resulting in a maxima being produced.

B1

OR

Where the path difference between the 2 waves at the receiver = $n\lambda$ ($n = 0, 1, 2, \dots$ and λ = wavelength of wave),

B1

constructive interference occurs and the two amplitudes add up to produce a maximum resultant amplitude, resulting in a maxima being produced.

B1

(ii) Movement of the plate from A to B produces an **extra path difference of $\frac{1}{2}$ wavelength**,

B1

destructive interference occurs and the two amplitudes cancel out to produce a minimum resultant amplitude, resulting in a minima being produced.

B1

(iii) **Amplitudes of the waves** reaching the receiver **are different** because they travel through different distances,

B1

Hence waves **do not cancel out completely**

B1

$$(iv) \quad \lambda = \frac{c}{f} = \frac{3.0 \times 10^8}{1.0 \times 10^9} = 0.300 \text{ m}$$

C1

$$2d = \frac{1}{2} \lambda$$

M1

$$\Rightarrow d = \frac{1}{4} \lambda = \frac{0.300}{4} = 0.075 \text{ m}$$

A1

- (b) (i) Incident sound wave travel up and gets reflected back at top of tube. B1

Incident and reflected waves which are of the same amplitudes, same frequency and speed travelling in opposite direction superpose to produce a standing wave.

B1

- (ii) Line has maximum value of amplitude at $h = 0$ cm and $h = 0.60$ m only

B1

Line has minimum/zero value of amplitude at $h = 0.30$ m only B1

- (iii) 1. vertical / along length or axis of tube B1

2. π or 3.14 rad B1

- (iv) On 1 half of a cycle, molecules moves towards (or approach) the node

B1

On next half of the cycle, molecules moves away from the node B1

Pressure will be changing from above to below atmospheric pressure.

B1

- (v) $c = f\lambda = f \times 2L = 275 \times 2 \times 0.60 = 330 \text{ ms}^{-1}$ M1

$$K = c^2 \times \rho = 330^2 \times 1.3 = 1.42 \times 10^5 \text{ kg m s}^{-1} \quad \text{A1}$$